Justification for the Spectral Loss Design: Wasserstein, KL-Divergence, and Soft-Max Peak Height

Wave Spectrum Forecasting with Transformers

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1 Context and Purpose

The model under development forecasts the directional wave spectrum $E(f,t)$ (or its unit-area normalized form, $E(f,t)/m_0$, the `shape` target) over a log-spaced frequency grid of 47 bins, between 0.02 and 0.485 Hz. Training and evaluation have so far relied on RMSE computed bin-by-bin between the predicted and observed spectrum. This document lays out the technical justification for complementing that RMSE term with three additional, physically motivated penalty terms — a Wasserstein-1 distance, a Kullback-Leibler (KL) divergence, and a differentiable soft-max peak-height term — all three already implemented in this project as `SpectralWassersteinLoss`, `SpectralKLDivergenceLoss`, and `SoftPeakHeightLoss` (`utils/loss.py`) — and situates that choice relative to how these mechanisms are already used in analogous domains of spectral/distributional comparison and differentiable extremum localization (audio, EEG, image retrieval, non-negative matrix factorization, human pose estimation).

2 Why RMSE Is a Poor Metric for Spectral Evaluation

RMSE between two spectra is defined bin by bin:

$$\mathrm{RMSE}(E_{\mathrm{pred}}, E_{\mathrm{true}}) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\big(E_{\mathrm{pred}}(f_i) - E_{\mathrm{true}}(f_i)\big)^2}. \tag{1}$$

This formulation treats each frequency bin as an independent, interchangeable coordinate: the error between f_i and f_{i+1} is computed exactly the same way as the error between f_i and f_{40}. The frequency axis — which has real, non-uniform, log-spaced geometry — is invisible to the metric. This produces three concrete problems for a wave spectrum:

1. **No notion of proximity between bins.** A peak predicted one bin off (a small, physically minor phase/position error) is penalized comparably to a peak that vanishes entirely or appears in a completely wrong frequency band (a physically severe energy/modality error). RMSE cannot distinguish these because there is no cost for *moving* energy mass from one bin to its neighbor — only a cost for it being in the right place or not.

2. **Structural bias toward smoothed spectra.** When there is genuine uncertainty about the exact position of the spectral peak across training samples (normal in ocean waves — the peak frequency f_p drifts hour to hour), the estimator that minimizes mean squared error is not "predict the most likely peak" but the *average* of several narrow spectra slightly shifted relative to one another. That average is broader and lower than any individual sample. This is the same mechanism, well documented, behind blur in image-generation models trained with pixelwise L_2/MSE loss.

3. **Insensitivity to modality errors.** In mixed sea states (wind sea + swell), the spectrum is bimodal or trimodal. A whole-spectrum RMSE dilutes exactly the error type that matters physically: getting total energy right while misplacing the relative amplitude or position of the two peaks, or collapsing two separate peaks into one.

In short: RMSE is well suited to *magnitude* accuracy per bin, but blind to *shape* and *position* — precisely what matters most in a wave spectrum, since total energy (proportional to H_s^2) is already tracked separately by the bulk-parameter metrics (`Hs_RMSE`, `Tm02_RMSE`) already computed in this pipeline.

3 The Wasserstein Distance as a Correction

The Wasserstein distance treats each spectrum (normalized to unit area, exactly what `compute_shape` produces) as a probability distribution over frequency, rather than a vector of independent coordinates. For two distributions μ, ν over the same metric space:

$$W_p(\mu,\nu) = \left(\inf_{\gamma \in \Pi(\mu,\nu)} \int |f-g|^p \, d\gamma(f,g) \right)^{1/p}, \tag{2}$$

where $\Pi(\mu,\nu)$ ranges over all transport plans moving the mass of μ onto ν. Intuitively, W_p is the minimal cost of physically moving energy from one part of the spectrum to another, where moving one unit of mass a given distance costs proportionally to that distance.

For one-dimensional distributions — exactly the case here, $E(f)/m_0$ over the frequency grid — W_1 has a closed form via cumulative distribution functions (CDFs):

$$W_1(\mu,\nu) = \int_0^\infty \left| F_\mu(f) - F_\nu(f) \right| df, \tag{3}$$

which is exactly what `SpectralWassersteinLoss` computes (via a discrete cumulative-trapezoid integral over the true grid).

This directly resolves the three problems from Section 2: a shifted peak costs little (proportional to how far it moved), an optimum under W_1 does not collapse to a blurred average the way an L_2-minimizing optimum does (translated copies of the same sharp bump have a non-blurred Wasserstein barycenter), and a modality error (mass moved from one peak to a distant one) is directly and heavily penalized because the transported distance is large.

4 Why Combine RMSE with a Wasserstein Penalty

The implemented approach does not replace RMSE — it composes:

$$\mathcal{L} = \mathrm{RMSE}\big(E_{\text{pred}}, E_{\text{true}}\big) \; + \; \lambda_W \cdot W_1\big(\hat{P}_{\text{pred}}, \hat{P}_{\text{true}}\big), \qquad \hat{P} = E/m_0. \tag{4}$$

Reasons to keep RMSE as the primary term rather than training on W_1 alone:

1. **Per-bin magnitude fidelity still matters.** $H_s = 4\sqrt{m_0}$ and T_{m02} are computed by direct integration of the spectrum; those integrals depend on point-wise magnitude, not only on the relative shape or position of energy mass. RMSE anchors that component.

2. W_1 **is defined on normalized distributions.** By construction it measures *shape* error, not total energy — comparing two unit-mass distributions discards scale information. This is desirable as a complementary term, but it cannot be the sole loss for a target that also needs absolute-magnitude accuracy.

3. **Training stability.** RMSE is smooth, strictly convex around the target, and well-behaved from the start of training, when predictions are still far from the target distribution.

4. λ_W **exposes the trade-off explicitly**, tunable via validation, between bin-wise fidelity and shape/position fidelity.

5 A Remaining Gap: Physically-Weighted Local Errors

RMSE and W_1 together still share one blind spot. Both treat every frequency bin as equally important *in isolation*: RMSE (computed in log-space, on $\log E(f)$ or $\log(E(f)/m_0)$, floored near zero to avoid $\log(0)$) assigns the same weight to a bin crossing that near-zero floor regardless of whether that bin is the spectral peak or an inconsequential tail bin; W_1's transport cost depends only on how far mass moved, not on how physically important the bin it moved from or to actually is.

This matters because the bulk parameters that ultimately matter — H_s, T_{m02} — come from integrals ($m_0 = \int E\,df$, $m_2 = \int f^2 E\,df$) that are dominated by the region around the spectral peak. A log-space error at a near-zero tail bin contributes almost nothing to these integrals; the identical log-space error at the peak can substantially distort them. Neither RMSE nor W_1, as currently combined, makes this distinction explicit — a further term is needed that weights each bin's error in proportion to how much of the spectrum's actual energy that bin represents.

6 KL Divergence as a Physically-Weighted Penalty

6.1 Definition and Derivation

Since $\hat{P} = E/m_0$ is already, by construction, a non-negative, unit-area probability distribution over frequency, the standard information-theoretic measure of divergence between two distributions applies directly — the Kullback-Leibler divergence:

$$D_{KL}(P \,\|\, Q) = \sum_i P(f_i) \, \log \frac{P(f_i)}{Q(f_i)}. \tag{5}$$

Expanding the logarithm, $\log \frac{P(f_i)}{Q(f_i)} = \log P(f_i) - \log Q(f_i)$ (no extraneous sign), and writing $\epsilon_i = \log P(f_i) - \log Q(f_i)$ for the per-bin *log-space error* — the same log-space representation the model already predicts and trains in:

$$D_{KL}(P \,\|\, Q) = \sum_i P(f_i) \cdot \epsilon_i. \tag{6}$$

Since $\sum_i P(f_i) = 1$, this is exactly a P**-weighted average** of the per-bin log-space errors — the weight on bin i is $P(i)$, the fraction of the true spectrum's total energy actually located there. Contrast with the plain log-space MSE currently used as the main per-bin loss:

$$\mathcal{L}_{\text{MSE-log}} = \frac{1}{N} \sum_i \epsilon_i^2, \tag{7}$$

where every bin carries the same weight, $1/N$, independent of position.

6.2 Non-Negativity and Relationship to Cross-Entropy

Individual terms $P(f_i)\cdot\epsilon_i$ can carry either sign — positive where Q underestimates P, negative where Q overestimates it — but the weighted sum is always non-negative, with equality if and only if $P = Q$ everywhere (Gibbs' inequality, a direct consequence of the concavity of log). This guarantees D_{KL} behaves as a proper training penalty, bounded below by zero. It is worth stating explicitly because expanding the logarithm the wrong way — introducing a spurious overall minus sign — yields $-D_{KL}(P\|Q)$, which is non-positive and would, if minimized, reward the optimizer for making the prediction worse rather than better; this is a one-symbol error easy enough to make that it is worth flagging as a check on the final implementation, not just the derivation.

D_{KL} is closely related to the cross-entropy $H(P,Q) = -\sum_i P(f_i)\log Q(f_i)$:

$$D_{KL}(P \,\|\, Q) = H(P,Q) - H(P), \tag{8}$$

where $H(P)$ is P's own Shannon entropy. Since $H(P)$ does not depend on the prediction Q, D_{KL} and $H(P,Q)$ produce *identical gradients* with respect to the predicted spectrum — this is exactly the log-softmax cross-entropy mechanism used in classification, applied over the frequency axis instead of over class labels, and it means the choice between the two is not a training-behavior decision. D_{KL} is preferred here purely for interpretability of the reported value: it is exactly zero at a perfect prediction, whereas plain cross-entropy floors at $H(P)$, a nonzero, sample-dependent value that makes "how close to perfect" harder to read off the raw loss.

6.3 Physical Interpretation for Bulk Wave Parameters

For the `shape` target, $P(f_i) = \big(E(f_i)/m_0\big)\cdot\Delta f_i$ is precisely the fraction of total wave energy carried by bin i. Any per-frequency log-space error is scaled by exactly how much of the spectrum's energy that bin represents: a tail bin (small $P(f_i)$) contributes almost nothing to D_{KL} even if its log-space error is large, while the same error at the spectral peak (large $P(f_i)$) is weighted heavily. This is not an arbitrary or hand-tuned weighting scheme — it falls directly out of treating the spectrum as a probability distribution, and it reproduces exactly the priority that H_s and T_{m02} already have physically, since both are dominated by the low-order moments of the energy-dense region around the peak.

The resulting composite loss:

$$\mathcal{L} = \mathrm{RMSE}\big(E_{\text{pred}}, E_{\text{true}}\big) \;+\; \lambda_W \cdot W_1\big(\hat{P}_{\text{pred}}, \hat{P}_{\text{true}}\big) \;+\; \lambda_{KL} \cdot D_{KL}\big(\hat{P}_{\text{true}} \,\|\, \hat{P}_{\text{pred}}\big) \tag{9}$$

gives each term ownership of a distinct failure mode: RMSE anchors absolute per-bin magnitude (needed for H_s/T_{m02} integration fidelity); W_1 penalizes energy transported across the frequency axis, including between distant spectral modes, in proportion to distance moved; D_{KL} penalizes per-bin log-space error in proportion to physical importance, correcting the one blind spot RMSE and W_1 share (Section 5) without reintroducing either of their own.

7 A Further Gap: Local Peak Sharpness

Even with D_{KL} added, RMSE, W_1, and D_{KL} together still share one remaining blind spot: none of them supplies a gradient signal specifically anchored to whether a detected spectral peak *retains its own amplitude.* W_1's transport cost is proportional to how far energy moves, not to how much a peak's own height changes — redistributing mass from a peak bin into its two immediate neighbors is geometrically cheap (mass moves only one or two bins) even though it can collapse the peak's amplitude by a large fraction. D_{KL}, as noted in `SpectralKLDivergenceLoss`'s own documentation, is dominated by whatever probability the prediction assigns exactly at the bins where the true distribution concentrates its mass — this makes it *decline to pile on* for a moderately broadened peak the way an unweighted MSE would, but it supplies no positive, targeted gradient pressure to restore the peak's amplitude once it has partially collapsed; it merely penalizes the collapse less harshly than MSE, rather than correcting it. Neither mechanism, individually or combined, behaves like a term whose purpose is specifically "does the peak's height survive."

This matters physically because spectral bandwidth and peakedness — closely tied to T_{m02} and to distinguishing a narrow, long-travelled swell partition from a broader, locally generated wind-sea partition — depend on whether the peak itself stays sharp, not merely on whether energy remains somewhere in its general neighborhood.

8 Soft-Max Peak Height as a Fourth Penalty Term

8.1 From Hard Maximum to a Differentiable Relaxation

Within a window $\mathcal{W}_k$ around a detected peak k, the natural summary of "the peak's height" is $\max_{i\in\mathcal{W}_k} E(f_i)$. As a training signal this is unusable: its gradient is exactly zero at every bin except the single arg-max bin, and that bin can flip discontinuously between optimization steps. Replacing the hard maximum with a temperature-controlled soft-max recovers a smooth, everywhere-differentiable surrogate:

$$\sigma_k(i;\tau) = \frac{\exp\big(E(f_i)/\tau\big)}{\sum_{j\in\mathcal{W}_k}\exp\big(E(f_j)/\tau\big)}, \qquad H_k^{\text{pred}}(\tau) = \sum_{i\in\mathcal{W}_k} E_{\text{pred}}(f_i)\,\sigma_k(i;\tau). \tag{10}$$

As $\tau\to\infty$, $\sigma_k\to$ uniform and $H_k^{\text{pred}}\to$ the window's plain mean; as $\tau\to 0$, $\sigma_k\to$ one-hot at the arg-max and $H_k^{\text{pred}}\to$ max. For any finite $\tau>0$ every bin in the window receives nonzero, informative gradient, weighted by its own current soft-max confidence.

8.2 Windows from the Existing Partition Detector

$\mathcal{W}_k$ is taken directly from the trough-to-trough partition boundaries already computed by this project's Portilla et al. (2009) significant-peak detector (`find_significant_peaks`), exposed for this purpose by a new `find_peak_windows` helper in `utils/spectral_partitioning.py` — narrow for a narrow swell partition, wide for a broad wind-sea partition, rather than an arbitrary fixed bin-radius that would necessarily be wrong for one of the two regimes.

8.3 Calibrating the Temperature

A window's width alone is not a safe basis for τ_k, for two reasons. First, the buoy's frequency grid is log-spaced and non-uniform (bins 0.005 Hz wide below 0.1 Hz, 0.02 Hz wide above 0.365 Hz), so the number of bins in a partition is not a reliable proxy for its physical bandwidth — the window width must be measured as $\Delta f_k = f(r_k) - f(\ell_k)$ in Hz, not as a bin count. Second, $\exp(E_i/\tau)$ is only numerically and conceptually meaningful when τ is expressed in E's own units (spectral density), not in frequency's — a temperature calibrated purely from bandwidth would make the soft-max's sharpness swing arbitrarily between a calm-sea sample and a storm sample with an identically *shaped* partition, since E's absolute scale carries no relation to a window's width in Hz. Combining both corrections, the temperature used is a fixed fraction of the peak's own (label-side) height, modulated by that peak's bandwidth relative to the grid's total span:

$$\tau_k = \max\left(\tau_{\min},\ c\cdot H_k^{\text{true}}\cdot\frac{f(r_k)-f(\ell_k)}{f_N - f_1}\right), \tag{11}$$

so a narrow swell partition gets a small, near-hard-max temperature, and a broad wind-sea partition gets a larger, softer one — matching the physical fact that a broad wind-sea peak genuinely does not have as sharply defined a height as a narrow swell peak. H_k^{true} is drawn only from the label, exactly as the window detection itself already is, so this is not a train/inference information leak.

8.4 Loss Definition

On the label side a hard maximum is used directly (no gradient is needed there): $H_k^{\text{true}} = \max_{i \in \mathcal{W}_k} E_{\text{true}}(f_i)$. The per-peak, then per-sample, loss is

$$\mathcal{L}_{\text{peak},k} = \left(H_k^{\text{pred}}(\tau_k) - H_k^{\text{true}}\right)^2, \qquad \mathcal{L}_{\text{peak}} = \frac{1}{K}\sum_{k=1}^{K} \mathcal{L}_{\text{peak},k}, \tag{12}$$

averaged over a sample's K detected peaks (samples with $K = 0$ are excluded from the batch average rather than contributing a filled-in zero, the same convention already used for undefined-quantity samples via `MO_MASK_THRESHOLD` in `nn/evaluate.py`). Adding this term gives the composite loss its fourth and final component:

$$\mathcal{L} = \text{RMSE}\big(E_{\text{pred}}, E_{\text{true}}\big) \; + \; \lambda_W \, W_1\big(\hat{P}_{\text{pred}}, \hat{P}_{\text{true}}\big) \; + \; \lambda_{KL} \, D_{KL}\big(\hat{P}_{\text{true}} \,\|\, \hat{P}_{\text{pred}}\big) \; + \; \lambda_{\text{peak}} \, \mathcal{L}_{\text{peak}}. \tag{13}$$

$\mathcal{L}_{\text{peak}}$ is implemented and unit-tested as `SoftPeakHeightLoss` in `utils/loss.py`, together with `find_peak_windows` in `utils/spectral_partitioning.py`; neither is yet wired into the training loop's hyperparameter search (the weight λ_{peak} is not yet an Optuna-tuned parameter), pending the same promotion path already followed by `SpectralWassersteinLoss` and `SpectralKLDivergenceLoss`.

9 Wasserstein, KL Divergence, and Soft-Max Localization in Analogous Fields

All three mechanisms are well established in other domains, for the same underlying reason: point-wise metrics ignore the geometry or relative importance of the axis being compared, and a hard extremum operator (max/ arg-max) carries no usable gradient at all — each of the three corrections addresses one of these structural gaps.

9.1 Wasserstein Distance

- **Image retrieval.** Rubner, Tomasi, and Guibas [1] introduced the Earth Mover's Distance as a similarity metric between image color histograms precisely because bin-wise L_1/L_2 distances penalize a color shift between adjacent bins as heavily as an unrelated color change.
- **Audio and music.** Logan and Salomon [2] compared musical timbre similarity by representing each track's spectrum as a Gaussian mixture and comparing mixtures via Earth Mover's Distance, since a direct Euclidean spectral comparison fails when the relevant spectral content is concentrated in slightly shifted frequency regions.
- **Biomedical signals / EEG.** Cazelles, Robert, and Tobar [3] define the "Wasserstein–Fourier distance" directly between power spectral densities of stationary time series — essentially the same problem addressed here, treating the normalized PSD as a probability distribution over frequency.

- **Signal processing, broadly.** Kolouri et al. [4] survey optimal-transport/Wasserstein methods across signal processing and machine learning, including spectral and waveform comparison.

- **Deep generative models.** Arjovsky, Chintala, and Bottou [5], in proposing the Wasserstein GAN, show formally that the Wasserstein distance stays continuous and gradient-informative even when two distributions have non-overlapping support — directly relevant to narrow, poorly aligned spectral peaks. Efficient computation is discussed by Cuturi [6] (Sinkhorn distances) and Peyré and Cuturi [7] (a general computational optimal transport reference), with a practical implementation available in POT [8].

9.2 KL Divergence and Weighted Spectral Divergences

- **Information theory foundations.** The divergence itself is due to Kullback and Leibler [9]; its role as the natural divergence underlying cross-entropy, and its relationship $D_{KL} = H(P, Q) - H(P)$, is standard material in information theory [10] and is the same mechanism underlying cross-entropy loss in classifier training in general [11].

- **Audio spectrogram factorization.** Lee and Seung [12] introduce a generalized KL-divergence cost function for non-negative matrix factorization, alongside the Euclidean one, precisely because it better respects the multiplicative, magnitude-sensitive structure of non-negative spectral data than a squared-error cost — the same motivation as Section 5. Févotte, Bertin, and Durrieu [13] extend this line of work with the closely related Itakura-Saito divergence for music spectrogram factorization, again choosing a divergence family that weights spectral errors by magnitude/importance rather than treating every bin uniformly, for the same physical reason argued here: a fixed absolute error matters far more where the signal carries real energy than where it does not.

9.3 Differentiable Soft-Max Localization

- **Human pose estimation.** Sun et al. [14] replace the standard, non-differentiable arg-max decoding of a heatmap-based keypoint detector with an "integral regression" that computes the joint coordinate as a soft-max-weighted expectation over the heatmap, precisely to obtain an end-to-end differentiable localization signal — the same replacement of max/arg-max by a soft-max-weighted average used here for spectral peak height.

- **Discrete relaxation, more generally.** Jang, Gu, and Poole [15] formalize the general principle that a temperature- controlled soft-max is a continuous relaxation of a discrete arg-max decision, recovering the hard decision as temperature $\to 0$ and a smooth, fully differentiable surrogate for any finite temperature — exactly the two limits (mean / hard-max) that τ_k interpolates between here.

- **Attention as soft-max pooling.** Vaswani et al. [16] popularize soft-max-weighted pooling as the core mechanism of attention; this project's own encoder already uses the same mechanism to collapse a spectrum's frequency bins into one embedding (`_FreqAttentionPool`, `nn/freq_embedding.py`) — the soft-max peak-height loss applies the identical mechanism as a loss term rather than as a model layer.

The recurring pattern across these domains — image histograms, audio spectra, EEG power spectral densities, non-negative spectrogram factorization, heatmap-based keypoint localization — is the same one motivating this project's loss design: when the compared object is fundamentally a *distribution of mass over a structured, physically meaningful axis* rather than a vector of independent coordinates, the geometry of that axis (Wasserstein), the relative importance of each location on it (KL divergence), and a smooth relaxation of any hard extremum operator on it (soft-max) each carry information that a plain point-wise metric structurally cannot represent.

10 Conclusion

Bin-wise RMSE is necessary but not sufficient for evaluating wave spectra: it anchors physical magnitude but has no notion of frequency-axis geometry or of how much any given bin actually matters, producing a known bias toward smoothed/blurred predictions and blindness to both modality errors and to systematically undervaluing peak accuracy relative to the spectral tail. The Wasserstein distance corrects the first gap by penalizing energy transport in proportion to distance; the KL divergence corrects the second by weighting each bin's error in proportion to the physical energy it represents — and, as Section 6 shows, is mathematically nothing more than a P-weighted average of the same log-space errors RMSE already computes, unweighted. A further gap remains even so (Section 7): neither mechanism supplies targeted gradient toward restoring a peak's own amplitude once it has locally collapsed into its neighbors, since W_1's cost tracks distance moved (small for a local collapse) and D_{KL} only declines to penalize it as harshly as MSE would, rather than correcting it. A differentiable soft-max peak-height term (Section 8) closes this gap directly, replacing the zero-gradient hard maximum with a temperature-controlled, everywhere-smooth relaxation whose temperature is itself calibrated from the peak's own physical bandwidth and energy scale. All three corrections are supported by extensive precedent in analogous domains facing the same structural problems. The resulting four-term loss

$$\mathcal{L} = \mathrm{RMSE} + \lambda_W\, W_1 + \lambda_{KL}\, D_{KL} + \lambda_{\text{peak}}\, \mathcal{L}_{\text{peak}}$$

gives each term ownership of a distinct, complementary failure mode rather than asking any single metric to cover all of them at once. `SpectralWassersteinLoss`, `SpectralKLDivergenceLoss`, and `SoftPeakHeightLoss` are all implemented and unit-tested in this project. `SpectralWassersteinLoss`'s weight is already an Optuna-tuned hyperparameter; `SpectralKLDivergenceLoss`'s weight is wired into the training loop as an opt-in term, pending a manual sweep before the same promotion; `SoftPeakHeightLoss` and its supporting `find_peak_windows` helper are implemented and tested in isolation but not yet wired into the training loop at all — that integration (pre-computing per-sample peak windows once, alongside `freq_means`/ `shape_means`, and threading a λ_{peak} weight through `train_one_epoch`) is the next step before this term can be trained against.

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